

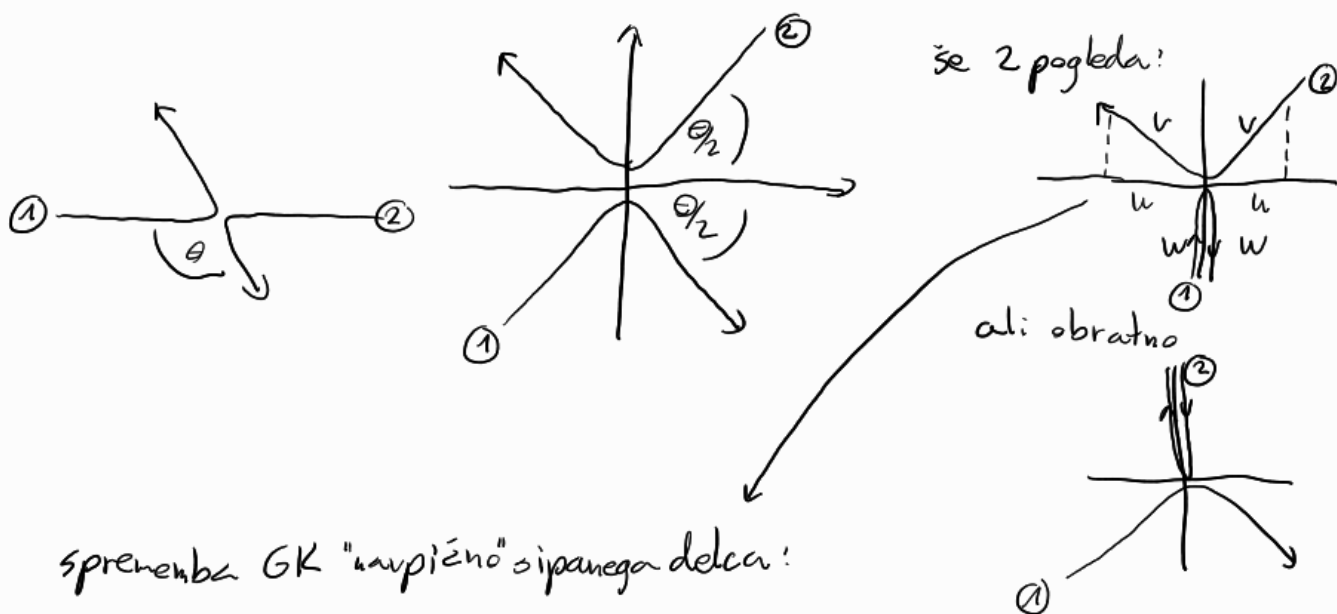
- $\vec{F} = m\vec{a}$ ne bo šlo, ker $\propto$ čas delovanja $\vec{F} \Rightarrow v \rightarrow \infty$

- Imamo že transf. posp.: $a'_x = \frac{a_x}{\gamma^3 (1 - \frac{u_x v}{c^2})^3} \dots$

→ najprej se lotimo GK:

- Predp., da ima obliko $\vec{p} = m_v \vec{v}$

Prožni trk identičnih delcev



sprememba GK "naprično" sipanega delca:

$$\Delta p = 2m_w w$$

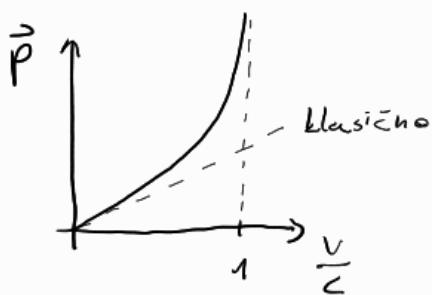
sprememba GK "póseвно" sipanega delca:

$$\Delta p' = 2m_v w \sqrt{1 - \beta_u^2}$$

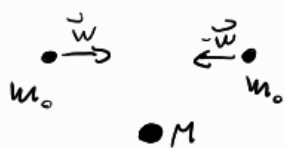
Želimo $\frac{\Delta p}{\Delta p'} = 1 \rightarrow \frac{m_w}{m_v} = \sqrt{1 - \beta_u^2} \rightarrow \lim_{w \rightarrow 0} \Rightarrow v \rightarrow u \Rightarrow \frac{m_0}{m_w} = \sqrt{1 - \beta_u^2}$

$\rightarrow \frac{m_0}{m_v} = \sqrt{1 - \beta_v^2} \rightarrow \boxed{m_v = \frac{m_0}{\sqrt{1 - \beta_v^2}}}$ (velja tudi v ne-limitnem primeru)
"relativistična masa"

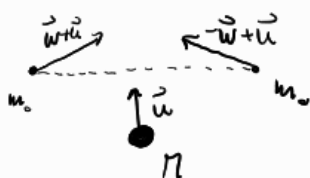
$\boxed{\vec{p} = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \vec{v} = m_0 \gamma \vec{v}}$ RELATIVISTIČNA GK



Nepr. trk (identičnih delcev)



pred trkom } gledamo v drigalu, ki gre $\vec{u}$
po trku } $\|\vec{u}\| \ll \|\vec{w}\|$



pred trkom }
po trku } $\rightarrow$ dirajiti se morajo navp. komp. GK:

$$p \approx 2m_w u \quad \text{pred trkom}$$

$$p' = M_0 u \approx M_0 u \quad \text{po trku}$$

$$\boxed{M_0 = 2m_w > 2m_0}$$

- masa sprimka > vsota mas zač. delcev

Kinetična energija

$$F = \frac{dp}{dt} = \frac{d(m_0 \gamma v)}{dt} \cdot \frac{dx}{dx}$$

$$F dx = m_0 v d(\gamma v) = m_0 v \gamma^3 dv$$

$$A = m_0 \int_{v=0}^v \gamma^3 v dv = -m_0 c^2 \int_{v=0}^v u^{-3} u du =$$

$$\left. \begin{aligned} u^2 &= \frac{1}{\gamma^2} = 1 - \frac{v^2}{c^2} \\ u du &= -\frac{v dv}{c^2} \end{aligned} \right\} \Rightarrow A = \boxed{E_k = m_0 c^2 (\gamma - 1)}$$

$$d(\gamma v) = d\gamma v + \gamma dv =$$

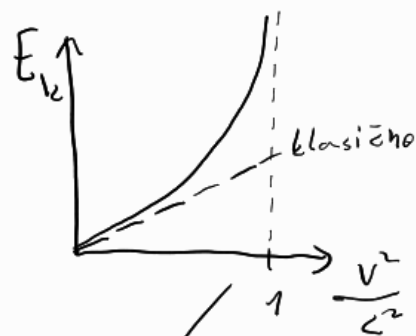
$$= + \underbrace{\frac{1}{\gamma^3} \left(1 - \frac{v^2}{c^2}\right)^{-3/2}}_{=\gamma^3} \left(\frac{2v dv}{c^2}\right) \cdot v + \gamma dv =$$

$$= \gamma^3 \frac{v^2}{c^2} dv + \gamma dv =$$

$$= \gamma^3 \left(\frac{v^2}{c^2} + \frac{1}{\gamma^2} \right) dv = \gamma^3 dv$$

$$\rightarrow \lim_{v \ll c} E_k = m_0 c^2 \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right) = m_0 c^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2} + \frac{3}{8} \frac{v^4}{c^4} \dots - 1 \right) =$$

$$= \frac{m_0 v^2}{2} + \dots$$



$$E_k = \underbrace{m_0 c^2 \gamma}_{\text{en. delca v gibanju (celotna en.)}} - \underbrace{m_0 c^2}_{\text{en. delca v mirovanju (mirovna energija)}}$$

za $\tilde{e}$ se to zgodi že pri
nekaj MeV ($U = \text{nekaj} \cdot 10^6 \text{ V}$)

$E_0 = m_0 c^2$	mirovna
$E = \gamma m_0 c^2$	celotna
$E_k = (\gamma - 1) m_0 c^2$	kinetična